# Companion to Bird & de Moor, Algebra of Programming

# §0. Laws from Relation Algebra

## §0.1. Adjunctions

(0.1a)

| | F⊣G | F's type | monad FG | 1⊑FG | GF⊑1 | F preserves ∪ | G preserves ∩ | FGF=F | GFG=G |
|---|---|---|---|---|---|---|---|---|---|
| 1 | ◁⊣▷ | A→<br>A⊗A | ◁▷=𝟙 | 𝟙⊑◁▷ | ▷◁⊑𝟙 | (R ∪ S)◁=<br>R◁ ∪ S◁ | (R∩S)▷=<br>R▷∩S▷ | ◁▷◁=◁ | ▷◁▷=▷ |
| 2 | ⊸⊣⊸ | A→𝕀 | ⊸⊸=⊤ | 𝟙⊑⊸⊸ | ⊸⊸⊑𝟙 | (R ∪ S)⊸=<br>R⊸ ∪ S⊸ | (R∩S)⊸=<br>R⊸∩S⊸ | ⊸⊸⊸=⊸ | ⊸⊸⊸=⊸ |
| 3 | °⊣° | (A→B)→<br>(B→A) | R°° = R | R°° = R | R°° = R | (R ∪ S)° =<br>S° ∪ R° | (R ∩ S)° =<br>R° ∩ S° | R°° = R | R°° = R |
| 4 | ⊸◁⊣▷⊸ | (X⊗A→Y)→<br>(X→Y⊗A) | 𝟙 | (⊸◁⊗𝟙)<br>(𝟙⊗▷⊸)=𝟙 | (𝟙⊗⊸◁)<br>(▷⊸⊗𝟙)=𝟙 | = | = | = | = |
| 5 | ▷⊸⊣⊸◁ | (X→Y⊗A)→<br>(X⊗A→Y) | 𝟙 | (𝟙⊗⊸◁)<br>(▷⊸⊗𝟙)=𝟙 | (⊸◁⊗𝟙)<br>(𝟙⊗▷⊸)=𝟙 | = | = | = | = |
| 6 | Δ⊣∩ | (A→B)→<br>(A→B)² | R ∩ R = R | R ∩ R = R | R ∩ S ⊑ R | Δ(R ∪ S)=<br>ΔR ∪ ΔS | (R∩T)∩(S∩U)=<br>(R∩S)∩(T∩U) | R ∩ R = R | R ∩ R = R |
| 7 | ∪⊣Δ | (A→B)²→<br>(A→B) | (R,S)↦<br>(R ∪ S,R ∪ S) | R ⊑ R ∪ S | R ∪ R = R | (R ∪ T) ∪ (S ∪ U)=<br>(R ∪ S) ∪ (T ∪ U) | Δ(R∩S)=<br>ΔR∩ΔS | R ∪ R = R | R ∪ R = R |
| 8 | 𝟘⊣! | {*}→<br>(A→B) | | | 𝟘 ⊑ R | | | | |
| 9 | ·S⊣/S | (A→B)→<br>(A→C) | S/S | R⊑(RS)/S | (T/S)S⊑T | (S ∪ T) R =<br>(S R) ∪ (T R) | (R ∩ R') / S =<br>(R / S) ∩ (R' / S) | ((RS)/S)S<br>=RS | ((T/S)S)/S<br>=T/S |
| 10 | S·⊣S\ | (B→C)→<br>(A→C) | S\S | R⊑S\(SR) | S(S\T)⊑T | R (S ∪ T) =<br>(R S) ∪ (R T) | S \ (R ∩ R') =<br>(S \ R) ∩ (S \ R') | S(S\(SR))<br>=SR | S\(S(S\T))<br>=S\T |
| 11 | R∩⊣R⇒ | (A→B)→<br>(A→B) | X↦R⇒(X∩R) | X⊑R⇒(X∩R) | R∩(R⇒Y)⊑Y | R∩(X ∪ Y)=<br>(R∩X) ∪ (R∩Y) | R ⇨ S ∩ T =<br>(R ⇨ S) ∩ (R ⇨ T) | R∩(R⇒(X∩R))<br>=X∩R | R⇒(R∩(R⇒Y))<br>=R⇒Y |
| 12 | 𝒟⊣·⊤ | (A→B)→<br>Cor A | R↦(𝒟R)⊤ | R⊑(𝒟R)⊤ | 𝒟(A⊤)⊑A | 𝒟(R ∪ S)=<br>𝒟R ∪ 𝒟S | (A∩B)⊤=<br>A⊤∩B⊤ | 𝒟((𝒟R)⊤)<br>=𝒟R | (𝒟(A⊤))⊤<br>=A⊤ |
| 13 | ℛ⊣⊤· | (A→B)→<br>Cor B | R↦⊤(ℛR) | R⊑⊤(ℛR) | ℛ(⊤A)⊑A | ℛ(R ∪ S)=<br>ℛR ∪ ℛS | ⊤(A∩B)=<br>⊤A∩⊤B | ℛ(⊤(ℛR))<br>=ℛR | ⊤(ℛ(⊤A))<br>=⊤A |
| 14 | ·f⊣·f° | (A→B)→<br>(A→C) | ff° | Map(f) ⇒ 𝟙 ⊑ f f° | f°f⊑𝟙 | (S ∪ T) R =<br>(S R) ∪ (T R) | Simple(g) ⇒<br>(X ∩ Y) g° =<br>(X g°) ∩ (Y g°) | ff°f=f | f°ff°=f° |
| 15 | f°·⊣f· | (A→C)→<br>(B→C) | ff° | Map(f) ⇒ 𝟙 ⊑ f f° | f°f⊑𝟙 | R (S ∪ T) =<br>(R S) ∪ (R T) | f(X∩Y)=<br>fX∩fY | ff°f=f | f°ff°=f° |
| 16 | i⊣E | Map↪Rel | E | $\frac{𝟙}{∋}$:A→PA | ∋:PB→B | — | — | $\frac{∋}{∋}$=𝟙 | $\frac{R}{∋}$∋=R |
| 17 | ·∋⊣ $\frac{}{∋}$ | Map(A,PB)→<br>(A→B) | 𝟙 | $\frac{f∋}{∋}$=f | $\frac{R}{∋}$∋=R | — | — | $\frac{f∋}{∋}$∋<br>=f∋ | $\frac{\frac{R}{∋}∋}{∋}$<br>=$\frac{R}{∋}$ |
| 18 | $\frac{}{∋}$ ⊣·∋ | (A→B)→<br>Map(A,PB) | 𝟙 | $\frac{R}{∋}$∋=R | $\frac{f∋}{∋}$=f | — | — | $\frac{\frac{R}{∋}∋}{∋}$<br>=$\frac{R}{∋}$ | $\frac{f∋}{∋}$∋<br>=f∋ |

## §0.2. Composing adjunctions

(0.2a)

| | | ·T | T· | ·g | g°· | Tn | ° | ⊶◁ | Δ |
|---|---|---|---|---|---|---|---|---|---|
| 1 | ·S | X / (S₁ S₂) = X / S₂ / S₁ | S \ (X / W) = (S \ X) / W | R/(Sg)= (Rg°)/S | Map(f) ⇒ f (R / S) = (f R) / S | — | (Y/S)°= S°\(Y°) | (RS)^= R^(S⊗𝟙) | (R ∩ R') / S = (R / S) ∩ (R' / S) |
| 2 | S· | S \ (X / W) = (S \ X) / W | (S₁ S₂) \ X = S₂ \ (S₁ \ X) | S\(Rg°)= (S\R)g° | (g°S)\R= S\(gR) | — | (S\Y)°= Y°/S° | ((S⊗𝟙)R)^= SR^ | S \ (R ∩ R') = (S \ R) ∩ (S \ R') |
| 3 | ·f | Map(f) ⇒ R / (f S) = (R / S) f° | T\(Rf°)= (T\R)f° | (R S)° = S° R° | | | (R S)° = S° R° | | Simple(g) ⇒ (X ∩ Y) g° = (X g°) ∩ (Y g°) |
| 4 | f°· | Map(f) ⇒ f (R / S) = (f R) / S | (Tf°)\R= f(T\R) | — | (R S)° = S° R° | | (R S)° = S° R° | | f(T₁∩T₂)= fT₁∩fT₂ |
| 5 | R∩ | — | — | — | — | R ⇨ S ⇨ T = R ∩ S ⇨ T | (R⇒Y)°= R°⇒(Y°) | — | R ⇨ S ∩ T = (R ⇨ S) ∩ (R ⇨ T) |
| 6 | ° | (Y/T)°= T°\(Y°) | (T\Y)°= Y°/T° | (R S)° = S° R° | (R S)° = S° R° | (T⇒Y)°= T°⇒(Y°) | R°° = R | (R^)°= (R°⊗𝟙) (𝟙⊗▷⊸) both bends: the ° section's display | (T₁∩T₂)°= T₁°∩T₂° |
| 7 | ⊶◁ | (RT)^= R^(T⊗𝟙) | ((T⊗𝟙)R)^= TR^ | — | — | — | (R^)°= (R°⊗𝟙) (𝟙⊗▷⊸) both bends: the ° section's display | — | — |
| 8 | ∪ | (S ∪ T) R = (S R) ∪ (T R) | R (S ∪ T) = (R S) ∪ (R T) | (S ∪ T) R = (S R) ∪ (T R) | R (S ∪ T) = (R S) ∪ (R T) | R ∩ (S ∪ T) = R ∩ S ∪ R ∩ T | (R ∪ S)° = S° ∪ R° | | |
| 9 | 0 | 0 R = 0 | R 0 = 0 | 0 R = 0 | R 0 = 0 | R ∩ 0 = 0 | 0° = 0 | | |

## §0.3. Adjoint triples

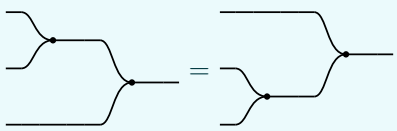
**(0.3a)**

```
·f⊣·f°⊣/f°
 f°·⊣f·⊣f\
```

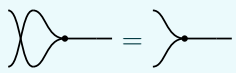
## §0.4. Frobenius

**Frobenius.**

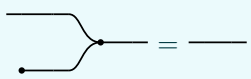
(0.4a)



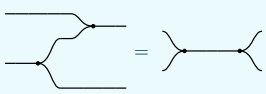
▷ associative



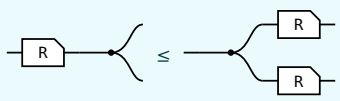
▷ commutative



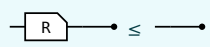
⊸ is its unit



Frobenius, one half — the other is its °

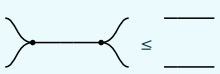



R ◁ ≤ ◁ (R ⊗ R)

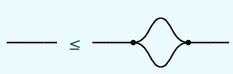



R ⊸ ≤ ⊸

(0.4b)



▷ ◁ ≤ 𝟙



𝟙 ≤ ◁ ▷



⊸ ⊸ ≤ 𝟙

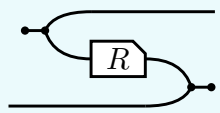


𝟙 ≤ ⊸ ⊸

## §0.5. ° is a contravariant 2-functor

**(0.5a)**

° is primitive, part of the data the first section lists. It turns both of R's wires round, and the Frobenius structure DRAWS that — the picture and the formula below are R°, not its definition:



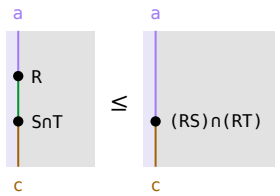

$$R^\circ=(\multimap\!\triangleleft\otimes\mathbb{1})(\mathbb{1}\otimes R\otimes\mathbb{1})(\mathbb{1}\otimes\triangleright\!\multimap)$$

where $\multimap\!\triangleleft : \mathbb{I}\to a\otimes a$ opens a pair of wires out of nothing and $\triangleright\!\multimap : b\otimes b\to\mathbb{I}$ closes one, so the input of $R^\circ$ is where the output of $R$ was. Taking that formula as the definition would now be circular: $\triangleleft$ and $\multimap$ are $\triangleright^\circ$ and $\multimap^\circ$. The laws of ° are that it is a contravariant 2-functor $^\circ : \mathcal{C}^{\circ\mathrm{p}}\to\mathcal{C}$:

(i) $\mathbb{1}^\circ=\mathbb{1}$ (ii) $(RS)^\circ=S^\circ R^\circ$ (iii) $(R\otimes S)^\circ=R^\circ\otimes S^\circ$ (iv) $R\le S$ implies $R^\circ\le S^\circ$

## §0.6. Meet and composition

(0.6a)

| semi-distributivity, and what supplies it | picture |
|---|---|
| R (S∩T)⊑RS∩RT — the lax copy law. Equality exactly when R is single valued: the Maps section's F(R∩S)=FR∩FS. |  |

## §0.7. Domain and range

(0.7a)

```
dom(R) ≜ 𝟙 ∩ (R R°)
```

```
Coreflexive(X) ⇒ dom(R) ⊑ X ⇔ R ⊑ X R
```

```
dom(R S) ⊑ dom(R)
```

```
dom(R ∩ S) = 𝟙 ∩ (S R°)
```

```
R entire Entire(R) ≜ dom(R) = 𝟙
```

```
R simple Simple(R) ≜ R° R ⊑ 𝟙
```

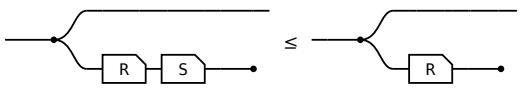
```
R a map Map(R) ≜ Entire(R) ∧ Simple(R)
```

```
Entire(R); Entire(S) ⇒ Entire(R S)
Simple(f); Simple(g) ⇒ Simple(f g)
Map(f); Map(g) ⇒ Map(f g)
```

```
Entire(R S) ⇒ Entire(R)
```

## §0.8. Sliding the discard

**(0.8a)**



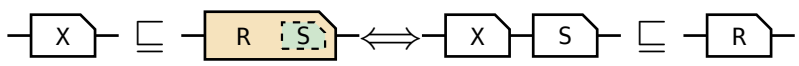

S⊸⊑⊸, the lax axiom for ⊸ in the first section — the discard slides back past S

## §0.9. Division

(0.9a)

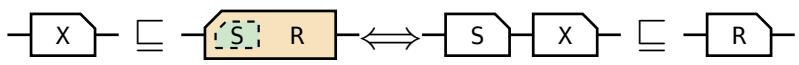
1 `X⊑R/S⇔XS⊑R`

X is any x-to-y pairing; one that only pairs x with a y such that x admires everyone y hates lies inside R/S, and R/S is the largest such.



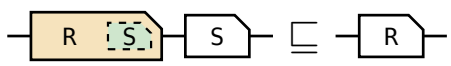

2 `X⊑S\R⇔SX⊑R`

The mirror — divide on the left when x comes first.



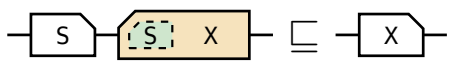

3 `(R/S)S⊑R`

There is a y such that x admires everyone y hates, and p is one of the people y hates — then x admires p too. Strict at S=∅: R/S is everyone, (R/S)S=∅.



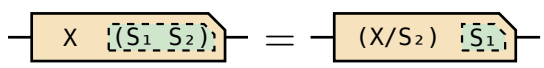

4 `S (S\R)⊑R`

The mirror.



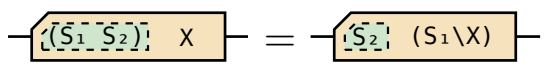

5 **associate:** `R/(S₁S₂)=(R/S₂)/S₁`

**A friend's enemy** is two hops: divide by the far end first.



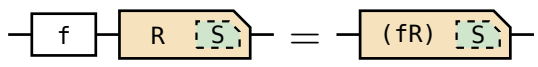

6 `(S₁S₂)\R=S₂\(S₁\R)`

The mirror.



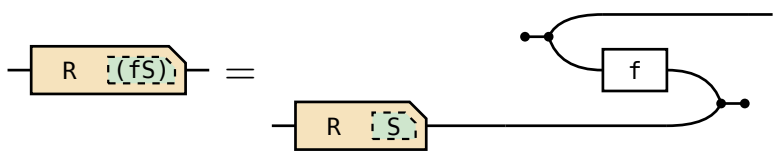

7 **maps:** `f (R/S)=(fR)/S`

Rename x before or after dividing — the licence to write fR/S.



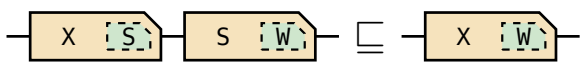

8 `R/(fS)=(R/S)f°`

Rename y: a map leaves a denominator as f° outside the box.



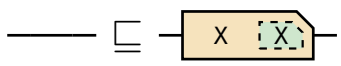

9 `(R/S)(S/W)⊑R/W`

Someone who admires all of a hate-set that already covers everyone z works for admires those people too.



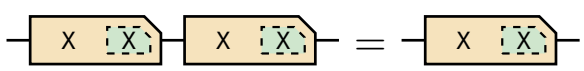

10 `𝟙⊑R/R`

R/R runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only a and b admire each other's idols too, and still stay two people.



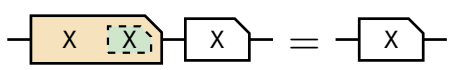

11 `(R/R)(R/R)=R/R`

R/R is the preorder **admires at least as much as**, and a preorder is idempotent. Freyd writes ⊑; with 𝟙⊑R/R above it is an equality.



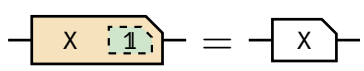

12 `(R/R)R=R`

Reaching p through someone whose idols x fully admires is reaching p directly, since x admires their own idols.



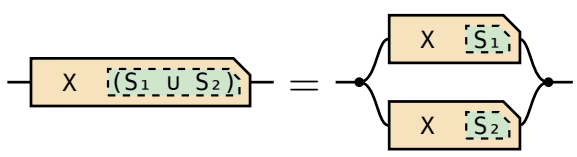

13 `R/𝟙=R`

Dividing by 𝟙: p's set is just {p}, so admiring all of it is admiring p.



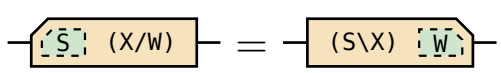

14 `R/(S₁ ∪ S₂)=R/S₁∩R/S₂`

Admiring a combined hate-set is admiring each set in full.




15 `S\(R/W)=(S\R)/W`

Which is why S\R/W needs no bracket.



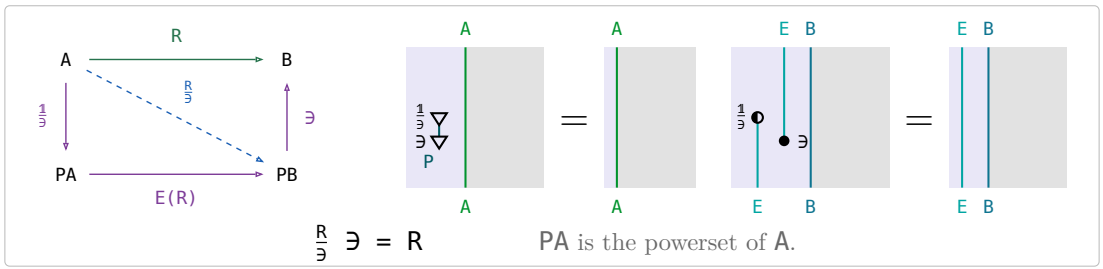
## §0.10. Power allegories

(0.10a)

| | law | the reading |
|---|---|---|
| 1 | $\ni_R \square = R\square$, $\ni_R = \ni_{R\,\square}$ | One ∋ per object, not per arrow. |
| 2 | ∋ is **thick** | **Comprehension**: every x has a set of exactly the people x admires. |
| 3 | $\frac{R}{\ni}$ : A→PB, for R : A→B | convert a relation to a function. a $\frac{R}{\ni}$ ={b\|a R b} |
| 4 | Map($\frac{R}{\ni}$) | |
| 5 | $\frac{R}{\ni}$ ∋ = R | reading the chosen set back through ∋ returns the relation. |
| 6 | Simple(F) ⇒ F ⊑ $\frac{F\,\ni}{\ni}$ | A partial choice of sets is inside the total one. |
| 7 | the **singleton map** is monic: $\frac{\mathbb{1}}{\ni}$ ($\frac{\mathbb{1}}{\ni}$)° ⊑ 𝟙 two points with the same one-person set are the same point. | The one-person set. |
| 8 | $\frac{\ni}{\ni}$ = 𝟙 | Make the set of a set, then read it back one level down. |
| 9 | **fusion:** Map(f) ⇒ $\frac{f\,R}{\ni}$ = f $\frac{R}{\ni}$ | Naturality of the unit, Map(f) ⇒ f $\frac{\mathbb{1}}{\ni}$ = $\frac{\mathbb{1}}{\ni}$ E(f) . renaming a point and then taking its one-person set is taking the set and renaming inside it. |
| 10 | Map(f); 𝟙□ = ∋□ ⇒ $\frac{f}{\ni}$ = f $\frac{\mathbb{1}}{\ni}$ | Rename first or take singletons first — the fusion row above at R=𝟙. |
| 11 | R□ = ∋□ ⇒ R /ₛ S = $\frac{R}{\ni}$ ($\frac{S}{\ni}$)° | x and y match when R sends x and S sends y to the same set. |
| 12 | E(R) ≜ $\frac{\ni\,R}{\ni}$ | E(R): PA→PB, image of a set of A |
| 13 | $\frac{R}{\ni}$ = $\frac{\mathbb{1}}{\ni}$ E(R) | $\frac{\mathbb{1}}{\ni}$: x↦{x} |
| 14 | $\frac{S}{\ni}$ E(R) = $\frac{S\,R}{\ni}$ | absorption — the monad's composition law, $\frac{S}{\ni}$ ⋄ $\frac{R}{\ni}$ = $\frac{SR}{\ni}$, §$\mathcal{A}$≅Kleisli(E) in Relation Algebra |
| 15 | E(R) = E($\frac{R}{\ni}$) union | the image of a set under R is the union of the R-sets of its points. |
| 16 | ⊇ ≜ ∋ / ∋ | xs⊇ys⇔∀a. ys∋a→xs∋a |
| 17 | ⊆ = ⊇° | xs⊆ys⇔ys⊇xs |
| 18 | T° $\frac{T}{\ni}$ ⊑ ∈ | every T-value of a point is a member of its T-set: Λ(T) is a map and Λ(T)∋=T. |

## §0.11. i⊣E: power transpose through singleton

(0.11a)

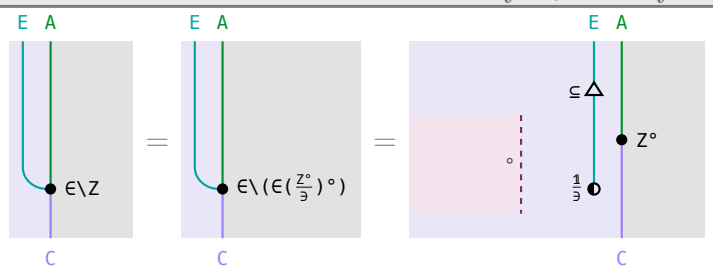



$\frac{R}{\ni}\ \ni = R$ PA is the powerset of A.

## §0.12. ∈\ as a composite

$$\in \setminus Z = \subseteq (\tfrac{Z^\circ}{\ni})^\circ \qquad \textbf{(0.12a)}$$

**xs is related to c by ∈\Z exactly when xs ⊆ Λ(Z°)(c), the set of every a with a Z c**




| | | |
|---|---|---|
| 1 | ∈ \ Z | Z=∈Λ(Z°)° — R/∋ = R |
| 2 | = ∈ \ (∈ (Z°/∋)°) | X\(Yf°)=(X\Y)f°, f a map |
| 3 | = ⊆ (Z°/∋)° | |